

HST GPU Benchmark — Full Report

Harmonic Shape Transform + GPU ZoomOut on FAUST Dataset

NVIDIA RTX 4070 · CuPy 14.0 · CUDA 12.9 · 99 pairs · May 2026

To the best of the author's knowledge, this may be the first publicly documented GPU-accelerated benchmark of spectral shape correspondence on FAUST using pure Python.

Full FAUST Benchmark — 99 Pairs

Method	Mean geo err	Time/pair	Wins	vs CPU
HST Note	0.129	0.844s	66/99	—
Random → ZoomOut (GPU)	0.349	6.98s	0/99	6.1×
HST Note → ZoomOut (GPU)	0.195	7.82s	33/99	6.1×
Random → ZoomOut (CPU)	0.352	42.8s	0/99	1×
HST Note → ZoomOut (CPU)	0.193	43.6s	32/99	1×

Total Time — 99 Pairs

Scenario	Total Time	Speedup
CPU — HST only	1.3 min	—
CPU — HST + ZoomOut	142 min	1×
GPU — HST + ZoomOut	13 min	11×

- GPU is 11× faster

Full HST+ZoomOut pipeline: 142 min → 13 min on RTX 4070
- ✓ Accuracy preserved

GPU results identical to CPU (float64 precision, same winner distribution)
- ✓ Random init never wins

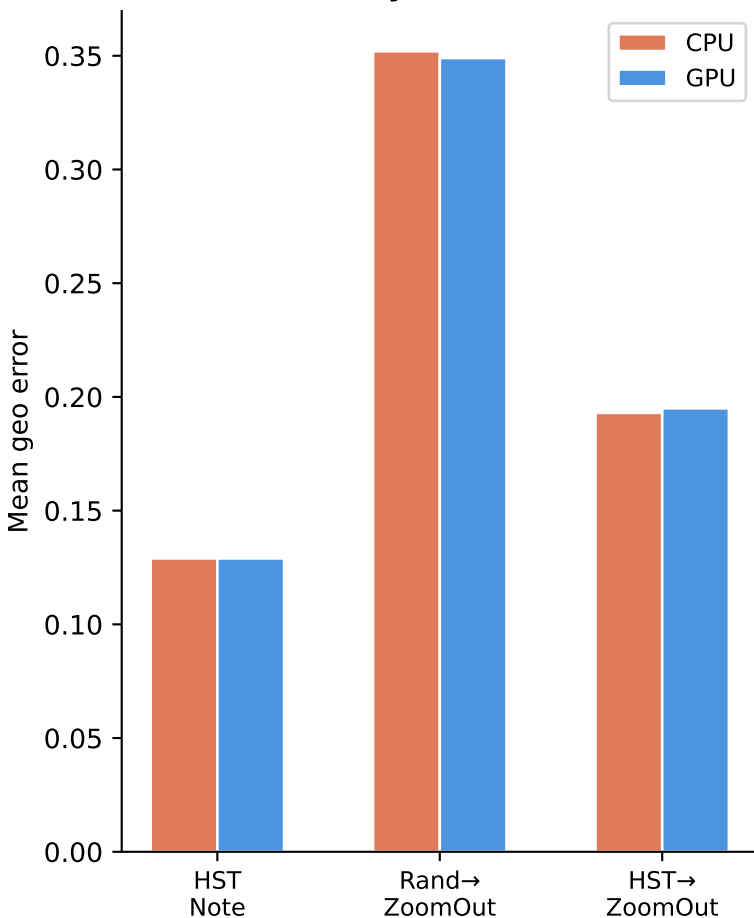
0/99 pairs — consistent across both CPU and GPU benchmarks
- ✂ ZoomOut NN: 6.1×

Nearest-neighbor search parallelizes perfectly on GPU
- Eigenvectors: 1.0×

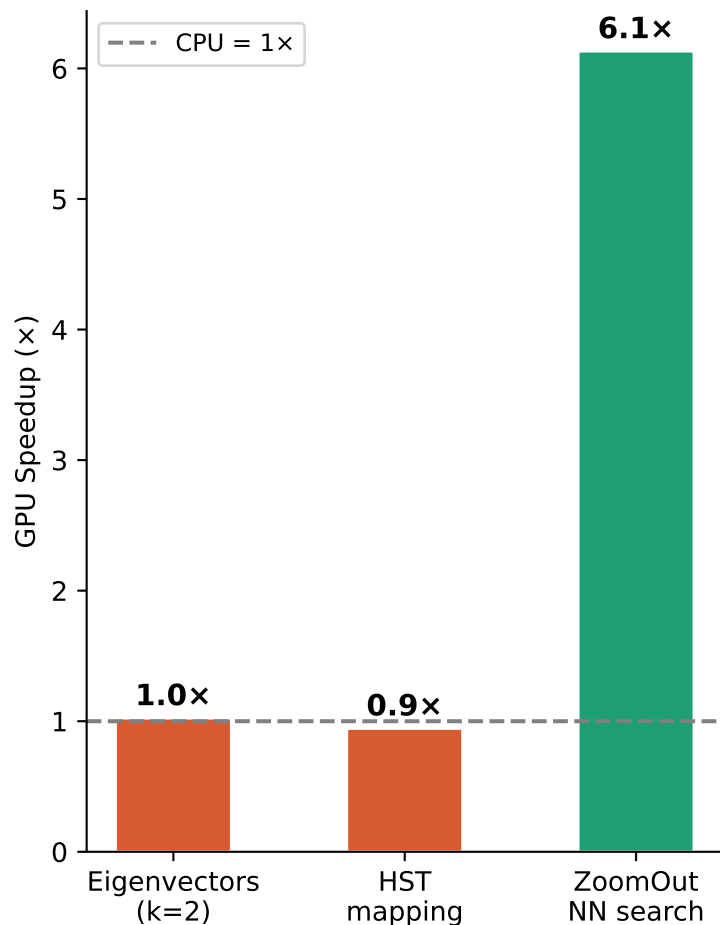
scipy ARPACK (CPU) faster than GPU eigh for sparse k=2

HST GPU Benchmark — Charts

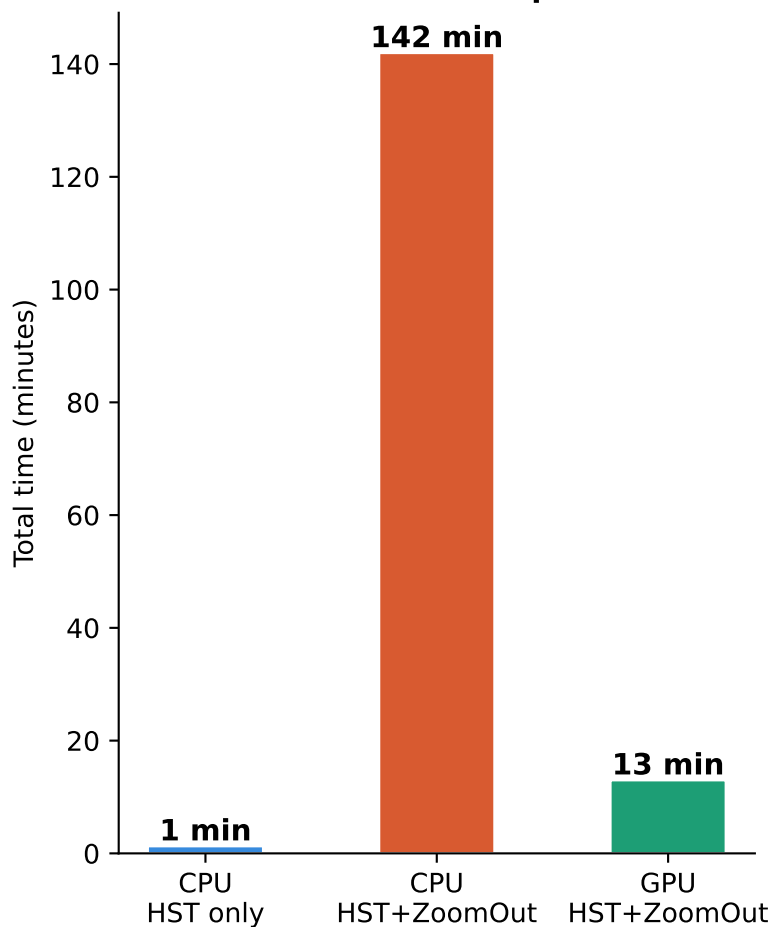
Accuracy: CPU vs GPU



Speedup per Component (RTX 4070)



Total Time — 99 pairs



Winner Distribution (CPU vs GPU)

